CALCULUS · UNIT 1: PRECALCULUS REVIEW

Phone Battery Lab

Review function families by modeling a phone's battery, charging, and app data.

CP1

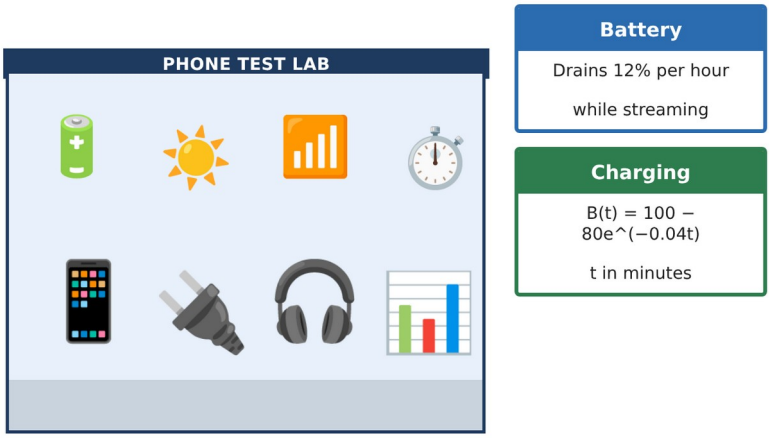
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A phone company wants a student lab to test its new phone. The battery drains while streaming, charges quickly at first and then slowly, the screen brightness follows the sunlight, and the data plan has a price that changes after a limit.

You will use the function families you need for calculus to model the phone and answer the company's questions.



Learning Goals

- I can build and interpret linear, exponential, logarithmic, and trig models.
- I can compose, invert, and transform functions.
- I can analyze end behavior and asymptotes.

Project Timeline

When	What you will do
Day 1	Drain & Charge
Day 2	Function Toolkit
Day 3	Brightness Cycle
Final	Battery Lab Report + reflection

Part 1: Drain & Charge

Day 1 · Use the Lab Data Cards.

Lab Data Cards

Streaming	Gaming	Charging
Starts at 95%	Loses 18% of the	B(t) = 100 − 80e^(−0.04t)
loses 12% per hour	remaining charge per hour	

Data Plan	Brightness	Temperature
$30 up to 15 GB	Max 900 nits at 1 p.m.	F = 1.8C + 32
+$4 per GB over	Min 100 nits at 1 a.m.	

1. When are the streaming (95 − 12t) and gaming ($95(0.82)^t$) batteries equal after t = 0? Use a graph or table.

2. Solve B(t) = 95 for the charging model.

3. Describe the end behavior of B(t) and write its horizontal asymptote.

4. Write the inverse of B(t) and explain what it tells you.

Part 2: Function Toolkit

Day 2 · Plans and conversions.

5. Write P(g) as a piecewise function and graph it. Is it continuous at g = 15?

6. Let C(F) = $\frac{F - 32}{1.8}$ and W(C) = 0.02C² (heat warning level). Find W(C(104)).

7. Describe the transformation from y = e^t to y = 100 − 80e^(−0.04t).

8. Find the average rate of change of B on [0, 30] and on [30, 60]. What do you notice?

Part 3: Brightness Cycle

Day 3 · Max 900 nits at 1 p.m. (t = 13), min 100 at 1 a.m.

9. Write a cosine model L(t) for brightness, t in hours after midnight.

10. Find L(8) and L(20).

11. When is brightness at least 700 nits?

Final Product: Battery Lab Report

Your final product must include:

- Comparison of drain models
- Charging inverse and asymptote
- Composition and continuity
- Brightness model and interval
- Lab report with graphs

Reflection

R1. What does the average rate of change tell you that a single value does not?

R2. Why is continuity of the data plan important to customers?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Models for battery, charging, plans, and brightness are built and used correctly. Goes beyond the minimum with a creative, realistic choice.	Models for battery, charging, plans, and brightness are built and used correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Lab report includes models, graphs, calculations, and conclusions. Polished, organized, and visually engaging.	Lab report includes models, graphs, calculations, and conclusions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.